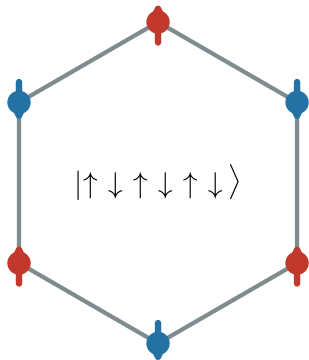
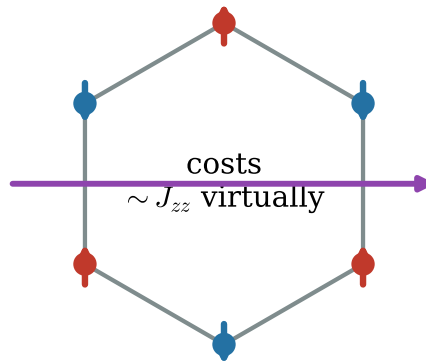


The physical ring exchange: amplitude $g_{\text{hex}} = 12|J_{\pm}|^3/J_{zz}^2$ (third order in $J_{\pm}$)

start (ice)



flip 3 alternating pairs



end (ice, flipped)

